

Student

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| Name: |  |
| ID:   |  |

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| Last Updated: |  |
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| General Education (22 credits)        |                                        | Cr. | Prerequisite                   | Co-requisite | Passing grade | Grade | points | validated credits | Remarks           | GE Course Codes         |
|---------------------------------------|----------------------------------------|-----|--------------------------------|--------------|---------------|-------|--------|-------------------|-------------------|-------------------------|
| ENG 240                               | English Communication                  | 3   |                                |              | 60            | 0     | 0      | 0                 |                   |                         |
| GEAD                                  | Artistic Discovery and Education       | 3   |                                |              | 60            | 0     | 0      | 0                 | Refer to GE shart | 200, 205, 210, 215, 220 |
| GERE                                  | Intercultural and Religious fluency    | 3   |                                |              | 60            | 0     | 0      | 0                 | Refer to GE shart | 200, 205, 210           |
| GEHS                                  | Lebanese History and Legacy            | 3   |                                |              | 60            | 0     | 0      | 0                 | Refer to GE shart | 205, 210, 215, 220      |
| ECO 350                               | Engineering Economics                  | 3   |                                |              | 60            | 0     | 0      | 0                 |                   |                         |
| GEN 410                               | Engineering Projects Management        | 2   | GEN350                         |              | 60            | 0     | 0      | 0                 |                   |                         |
| GEN 303                               | Innovation&Entrep.for Engineer         | 2   | GIN480                         |              | 60            | 0     | 0      | 0                 |                   |                         |
| GEN 301                               | Law for Engineers                      | 2   | Language                       |              | 60            | 0     | 0      | 0                 |                   |                         |
| GEN 302                               | Engineering Ethics                     | 1   | Language                       |              | 60            | 0     | 0      | 0                 |                   |                         |
| Mathematics and Sciences (32 credits) |                                        | Cr. | Prerequisite                   | Co-requisite | Passing grade |       |        |                   |                   |                         |
| CHM 212                               | General Chemistry                      | 3   |                                |              | 60            | 0     | 0      | 0                 |                   |                         |
| CHM 270                               | General Chemistry Lab                  | 1   |                                | CHM212       | 60            | 0     | 0      | 0                 |                   |                         |
| GEN 250                               | Modern Physics                         | 3   | MAT220                         | GEN270       | 60            | 0     | 0      | 0                 |                   |                         |
| GEN 270                               | Physics Lab                            | 1   |                                |              | 60            | 0     | 0      | 0                 |                   |                         |
| GEN 350                               | Mathematics for Engineers              | 3   | MAT220, MAT310, MAT313         |              | 60            | 0     | 0      | 0                 |                   |                         |
| GEN 428                               | Numerical Analysis                     | 3   | MAT220, MAT310, GIN221, STA320 |              | 60            | 0     | 0      | 0                 |                   |                         |
| MAT 202                               | Elements of Mathematical Structures    | 3   |                                |              | 60            | 0     | 0      | 0                 |                   |                         |
| MAT 213                               | Single Variable Calculus               | 3   |                                |              | 60            | 0     | 0      | 0                 |                   |                         |
| MAT 220                               | Differential Equations                 | 3   | MAT213                         |              | 60            | 0     | 0      | 0                 |                   |                         |
| MAT 310                               | Linear Algebra                         | 3   | MAT202                         |              | 60            | 0     | 0      | 0                 |                   |                         |
| MAT 313                               | Multivariable Calculus                 | 3   | MAT213                         |              | 60            | 0     | 0      | 0                 |                   |                         |
| STA 320                               | Probability and Statistics             | 3   | MAT213                         |              | 60            | 0     | 0      | 0                 |                   |                         |
| Engineering courses (92 credits)      |                                        | Cr. | Prerequisite                   | Co-requisite | Passing grade |       |        |                   |                   |                         |
| GEL 211                               | Electric Circuits                      | 3   | MAT220(Y)                      | GEL271       | 60            | 0     | 0      | 0                 |                   |                         |
| GEL 271                               | Electric Circuits Lab                  | 1   |                                | GEL211       | 60            | 0     | 0      | 0                 |                   |                         |
| GEL 311                               | Logic Design                           | 3   | GIN221                         |              | 70            | 0     | 0      | 0                 |                   |                         |
| GEL 313                               | Electronics                            | 3   | GEL211                         | GEL371       | 70            | 0     | 0      | 0                 |                   |                         |
| GEL 314                               | Digital Electronics                    | 2   | GEL311                         | GEL372       | 70            | 0     | 0      | 0                 |                   |                         |
| GEL 371                               | Electronics Lab                        | 1   | GEL271                         | GEL313       | 70            | 0     | 0      | 0                 |                   |                         |
| GEL 372                               | Digital Circuits Lab                   | 1   |                                | GEL314       | 70            | 0     | 0      | 0                 |                   |                         |
| GEL 420                               | Non Linear Electronics                 | 3   | GEL313                         | GEL472       | 70            | 0     | 0      | 0                 |                   |                         |
| GEL 445                               | Microprocessors                        | 3   | GEL314                         |              | 70            | 0     | 0      | 0                 |                   |                         |
| GEL 472                               | Non Linear Electronics Lab             | 1   |                                | GEL420       | 70            | 0     | 0      | 0                 |                   |                         |
| GEL 474                               | Microprocessors Lab                    | 1   | GEL445                         |              | 70            | 0     | 0      | 0                 |                   |                         |
| GEN 499                               | Seminars and Conferences               | 0   | GIN596                         |              | P             |       |        |                   |                   |                         |
| GIN 221                               | Introduction to Programming            | 3   |                                |              | 60            | 0     | 0      | 0                 |                   |                         |
| GIN 231                               | Data Structures and Algorithms         | 3   | GIN221                         |              | 60            | 0     | 0      | 0                 |                   |                         |
| GIN 300                               | Database Systems                       | 3   | GIN231                         | GIN371       | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 311                               | Elements of Discrete Mathematics       | 2   | MAT202, GIN221                 |              | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 314                               | Object Oriented Design                 | 3   | GIN231                         |              | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 321                               | Algorithmics                           | 3   | GIN231, GIN311                 |              | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 371                               | Database Lab                           | 1   |                                | GIN300       | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 400                               | Advanced Database Systems              | 3   | GIN300, GIN321                 | GIN401       | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 401                               | Advanced Database Lab                  | 1   |                                | GIN400       | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 421                               | Operating Systems                      | 3   | GIN321                         |              | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 425                               | Software Engineering Design            | 3   | GIN314                         |              | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 446                               | Web Programming                        | 3   | GIN300                         |              | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 450                               | Advanced Computer Architecture         | 3   | GEL445                         |              | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 480                               | Internship I                           | 1   | GIN446                         |              | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 527                               | Distributed Systems                    | 3   | GRT431                         |              | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 581                               | Internship II                          | 1   | GIN480                         | GIN596       | 70            | 0     | 0      | 0                 |                   |                         |
| GIN 596                               | Final Project I                        | 1   | Special Approval               |              | P             | 0     | 0      | 0                 |                   |                         |
| GIN 597                               | Final Project II                       | 3   | GIN596                         |              | 70            | 0     | 0      | 0                 |                   |                         |
| GRT 431                               | Network Architecture and Protocols     | 3   | GEL311, GIN231                 | GRT473       | 70            | 0     | 0      | 0                 |                   |                         |
| GRT 473                               | Network Architecture and Protocols Lab | 1   |                                | GRT431       | 70            | 0     | 0      | 0                 |                   |                         |
|                                       | Approved Faculty Electives             | 6   |                                |              | 60/70         |       |        | 0                 |                   |                         |
|                                       | Technical electives                    | 17  |                                |              | 70            |       |        | 0                 |                   |                         |
|                                       | TOTAL                                  | 143 |                                |              |               |       |        | 0                 |                   | 0                       |

| Technical Electives |                                                        | Cr. | Prerequisite    | Co-requisite | Passing grade |   |   |   |  |
|---------------------|--------------------------------------------------------|-----|-----------------|--------------|---------------|---|---|---|--|
| GEL 521             | Machine Learning Control                               | 2   | GEN428, GIN321  |              | 70            | 0 | 0 | 0 |  |
| GIN 525             | Computer Network Security                              | 3   | GRT431          |              | 70            | 0 | 0 | 0 |  |
| GIN 526             | Planning and Configuration of Computer Networks        | 3   | GRT431          |              | 70            | 0 | 0 | 0 |  |
| GIN 528             | Mobile Devices Programming                             | 2   | GIN446          |              | 70            | 0 | 0 | 0 |  |
| GRT 531             | Advanced Networks Architectures                        | 3   | GRT431          | GRT532       | 70            | 0 | 0 | 0 |  |
| GRT 532             | Advanced Networks Architectures Lab                    | 1   |                 | GRT531       | 70            | 0 | 0 | 0 |  |
| GEL 558             | Microcontrollers                                       | 3   | GEL445, GIN231  | GEL575       | 70            | 0 | 0 | 0 |  |
| GEL 575             | Microcontrollers Lab                                   | 1   | GEL474          | GEL558       | 70            | 0 | 0 | 0 |  |
| GRT 560             | Digital Image Processing and Coding                    | 3   | GRT410, STA320  |              | 70            | 0 | 0 | 0 |  |
| GRT 573             | Digital Image Processing Lab                           | 1   | GRT560 (Y)      |              | 70            | 0 | 0 | 0 |  |
| GIN 515             | Deep Learning                                          | 3   | GEL521          |              | 70            | 0 | 0 | 0 |  |
| GEN 510             | Agile Methodology: Design&Dev.                         | 2   | GIN221, GEN 410 |              | 70            | 0 | 0 | 0 |  |
| GIN 456             | Human-Robot Interaction: Design Principles and Methods | 3   | GIN446          |              | 70            | 0 | 0 | 0 |  |
|                     | Pre-approved 500 level engineering course              | 3   |                 |              | 70            | 0 | 0 | 0 |  |

| Approved Faculty Electives |                              | Cr. | Prerequisite | Co-requisite | Passing grade |  |   |   |  |
|----------------------------|------------------------------|-----|--------------|--------------|---------------|--|---|---|--|
| GMC 320                    | Dynamics                     | 3   | MAT220       |              | 60            |  | 0 | 0 |  |
| GRT 410                    | Signals and Systems          | 3   | GEN350       |              | 70            |  | 0 | 0 |  |
| GMC 340                    | Thermodynamics               | 3   | CHM212       |              | 70            |  | 0 | 0 |  |
| GRT 423                    | Waves and Propagation        | 3   | GRT320       |              | 70            |  | 0 | 0 |  |
| GEN 450                    | Finite Elements              | 3   | GEN428       |              | 70            |  | 0 | 0 |  |
| GRT 320                    | Electrostatics and Magnetism | 3   | MAT313       |              | 70            |  | 0 | 0 |  |